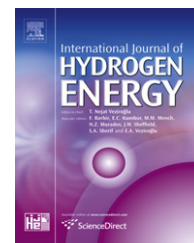


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Experimental study of hydrogen storage with reaction heat recovery using metal hydride in a totalized hydrogen energy utilization system

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ABSTRACT

Experimental results for hydrogen storage tanks with metal hydrides used for load leveling of electricity in commercial buildings are described. Variability in electricity demand due to air conditioning of commercial buildings necessitates installation of on-site energy storage. Here, we propose a totalized hydrogen energy utilization system (THEUS) as an on-site energy storage system, present feasibility test results for this system with a metal hydride tank, and discuss the energy efficiency of the system. This system uses a water electrolyzer to store electricity energy via hydrogen at night and uses fuel cells to generate power during the day. The system also utilizes the cold heat of reaction heat during the hydrogen desorption process for air conditioning. The storage tank has a shell-like structure and tube heat exchangers and contains 50 kg of metal hydride. Experimental conditions were specifically designed to regulate the pressure and temperature range. Absorption and desorption of 5,400 NL of hydrogen was successfully attained when the absorption rate was 10 NL/min and desorption rate was 6.9 NL/min. A 24-h cycle experiment emulating hydrogen generation at night and power generation during the day revealed that the system achieved a ratio of recovered thermal energy to the entire reaction heat of the hydrogen storage system of 43.2% without heat loss.

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1. Introduction

1.1. Background

Air conditioning in commercial buildings in Japan not only increases electrical power demand but also increases the demand gap between daytime and nighttime. Load leveling of electrical power demands saves energy cost and reduces the

environmental impact by “leveling” these demand variations and by depending more on base-load power plants because such plants generally produce less CO₂. Energy demand for air conditioning, especially in commercial buildings, must be stored during off-peak hours.

Hydrogen is a promising candidate as an energy storage medium due to its storability and portability [1]. Water electrolyzers produce hydrogen from water during off-peak hours

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and fuel cells supply electric power during peak hours to level out the daily variations in power demand. Compared with compressed hydrogen or liquid hydrogen in vessels, hydrogen storage with metal hydrides is suitable for stationary on-site storage due to its relatively low operating pressure and moderate operating temperature. These advantages plus a high volumetric storage density compensate for disadvantages of using metal hydrides, such as its heaviness. Bossi et al. used a metal hydride tank to examine power generation with a fuel cell during fluctuating demands [2].

Because the hydrogen absorption/desorption process of metal hydrides is an exothermic/endothermic process, thermal management of metal hydrides is essential for achieving a continuous process. Metal hydride in a reaction vessel is granular and its effective thermal conductivity is small, $\approx 1 \text{ W/(mK)}$. The equilibrium pressure of metal hydrides depends on temperature, and an efficient heat exchanger is required to maintain a continuous reaction and stable hydrogen flow. Heat and mass transfer in a metal hydride tank have been investigated experimentally and numerically. Tubular-type heat exchangers have been applied to hydrogen storage systems with metal hydride reactors [3–13]. Mellouli et al. studied a spiral tube in a reactor to enhance its heat exchange ability [13]. Botzung et al. validated the concept of a combined heat and power system and used a metal hydride tank to store excess hydrogen and to deliver it during high consumption periods [14]. Metal foam is also employed to smooth the distribution of the reaction heat during the absorption and desorption [15–17]. To evaluate reaction heat of metal hydride and heat exchange between the metal hydride and the heat transfer media such as coolant or air, various mathematical models and numerical simulation of heat and mass transfer in metal hydride beds have been proposed [18–23].

As reviewed by Muthukkumar and Groll [24], metal hydride has also been used in a heat pump system to store thermal energy [25–27]. These heat pump systems use two or more types of metal hydrides with different physical properties, and transport hydrogen between these metal hydrides to achieve reaction heat of absorption or desorption without continuous hydrogen production or consumption. Koseki et al. designed large-scale metal hydrogen tanks for a heat pump system to examine the pressure profile during heat pump operation [27].

Studies on applications of metal hydrides can be categorized into two groups: 1) hydrogen storage for other system, such as water electrolyzer and fuel cell, and 2) heat pumps using reaction heat generated by transferring hydrogen between two or more metal hydride tanks. There has been relatively little attempt to utilize both hydrogen storage and reaction heat. In our original study, first we introduce the concept of our totalized hydrogen energy utilization system (THEUS), in which a metal hydride tank absorb and desorb hydrogen for the storage and simultaneously utilize reaction heat for cold energy demand. Then, we present experimental results from a feasibility study on a metal hydride tank to store both hydrogen itself and thermal energy as an important component of THEUS. Finally, we evaluate the efficiency of energy storage of the metal hydride tank and its potential for load leveling.

1.2. Totalized hydrogen energy utilization system

An energy storage system that uses hydrogen has been previously proposed and is called the “Totalized Hydrogen Energy Utilization System (THEUS).” [28] The system involves water electrolyzers, metal hydride tanks, and fuel cells. The water electrolyzers and fuel cells can be replaced by a unitized reversible fuel cell proposed by Kato et al. [29] THEUS has a potential to be installed on-site at commercial buildings in Japan.

Fig. 1 illustrates a conceptual overview of THEUS. This system stores the electricity energy at night by using a water electrolyzer and generates power during the day by using fuel cells. The system also utilizes the cold heat of reaction heat for air conditioning. At night, THEUS produces hydrogen by using a water electrolyzer powered by inexpensive nighttime electrical power and then stores hydrogen by a metal hydride tank. During the day, THEUS generates electrical power by using fuel cells with the stored hydrogen to meet power demands in the respective building. A local system such as THEUS takes advantage of a simultaneously generated cold source and heat source to improve total energy efficiency. The metal hydride tank of THEUS does not store hydrogen only for the fuel cells of THEUS. Stored hydrogen can be used for fuel cell vehicles. THEUS can (1) optimizes the integration of energy storage and poly-generation of electric power, cold source, heat source and hydrogen to meet energy demand, (2) balances the load leveling, and (3) store renewable energy such as solar energy and wind energy. THEUS can bridge the gap between renewable energy and energy demand. Use of renewable energy sources becomes easier when THEUS is installed because fluctuations in electrical power loads can be managed. In this study, we focus on the metal hydride tank in THEUS to experimentally examine efficient operating conditions.

1.3. Operation principles of metal hydride tanks in THEUS

To ensure absorption/desorption ability of metal hydride for load leveling and to maximize the thermal efficiency of the total system, the operating conditions must be designed carefully. Table 1 shows the system requirements for the tank

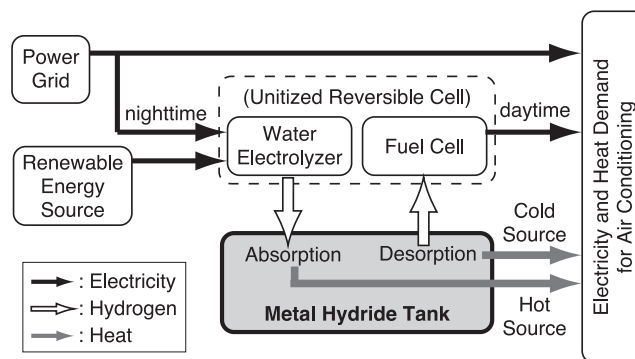


Fig. 1 – Concept of Totalized Hydrogen Energy Utilization System (THEUS) showing flow of electricity, hydrogen, and heat.

Table 1 – System requirements for the metal hydride tank.

Connected to	Absorption (Nighttime)	Desorption (Daytime)
	Water Electrolyzer	Fuel Cell
Time	13 h	9 h
Pressure	$P < 1.0 \text{ MPa}$	$P > 0.1 \text{ MPa}$
Inlet temperature	$T_{\text{in}} \leq 32 \text{ }^{\circ}\text{C}$	$T_{\text{in}} \geq 12 \text{ }^{\circ}\text{C}$
Outlet temperature	$32 \text{ }^{\circ}\text{C} \leq T_{\text{out}} \leq 35 \text{ }^{\circ}\text{C}$	$7 \text{ }^{\circ}\text{C} \leq T_{\text{out}} \leq 12 \text{ }^{\circ}\text{C}$

vessel during load leveling operation. We assume that during a 24-h load leveling cycle, there are 9 h of absorption and 13 h of desorption with an hour interval between the two processes. The operating pressure of the metal hydride must be carefully considered in Japan. According to the High Pressure Gas Safety Act in Japan, the maximum pressure during the absorption process must be less than 1.0 MPa(abs). Pressure during the desorption process must be higher than atmospheric to transfer hydrogen from the tank to the fuel cell without using a pump.

During the absorption process, the exothermic reaction heat must be exhausted from the metal hydride bed to the environment via coolant in order to maintain the metal hydride temperature and pressure. Assuming the exhaust heat is emitted by using conventional cooling towers, the coolant temperature from the outlet during the absorption process should be high enough to exchange heat even during a hot summer night. In this study, the cooling tower was assumed to work well when the coolant temperature was over 32 °C and to return the 32 °C coolant to the tank inlet. During the desorption process, on the other hand, the endothermic reaction heat can be utilized to cool the working fluid in refrigerators when the coolant is cold enough. In this study, the coolant was supplied to a refrigerator system only when the coolant temperature was less than 12 °C and then the 12 °C coolant was returned to the tank inlet. In our actual experiment, temperature control units adjusted the coolant temperature depending on the process (absorption/desorption) to emulate the external systems such as cooling towers and refrigerators. Hydrogen absorption with hot coolant and hydrogen desorption with cold coolant are difficult to achieve, compared to ordinary hydrogen storage systems that involve absorption with cold coolant and desorption with hot coolant.

The suitable mass flow rate of coolant (G_c) for the desired temperature difference (ΔT) can be calculated from the hydrogen flow rate (G_H) and the energy balance equation between reaction heat and coolant as follows:

$$G_H \Delta H = G_c (\rho c)_c \Delta T \quad (1)$$

where ΔH is the enthalpy of hydrogen absorption and $(\rho c)_c$ is the thermal capacity of the coolant. In this study, tap water with antifreeze solution is used as coolant. In following sections, the coolant is described as circulation water.

2. Experimental system

The overall experimental system used in this study consisted of a metal hydride tank, temperature control system, and

hydrogen supply (Figs. 1 and 2). The hydrogen supply played the role of a water electrolyzer. Any fuel cell was not connected to hydrogen outlet of the tank. Desorbed hydrogen is emitted to the outer environment. Actual load scenarios were emulated using a temperature control system to emulate the refrigerator and the cooling tower in an ideal situation.

2.1. Metal hydride tank

Fig. 2 shows a photograph (a) and schematic (b) of the metal hydride tank used in this study. The tank was made of a stainless steel cylindrical pipe and flanges, and had an internal diameter of 297.9 mm, an internal height of 346 mm, and a wall thickness of 10.3 mm. Metal hydrides expand in volume during the absorption process, and thus cause plastic deformation of stainless steel vessels [30,31]. To avoid this deformation, the metal hydride bed used in this study were separated and isolated from the outer wall. The hydride bed was separated into 10 compartments by 9.5-mm-thick copper plates to reduce the stress caused by expansion of the metal hydride. The copper plates also help dissipate the heat in the metal hydride bed. Stainless steel rings (inner diameter of 254.6 mm and thickness of 29.5 mm) were attached to the inner wall of the tank with rubber rings to reduce both the stress and heat applied to the tank. The inner volume of the tank was 24.1 L, and thus not larger than class-2 pressure vessels as regulated by the Industrial Safety and Health Act in Japan. The net volume of the void space in the tank was 12.96 L before being packed with metal hydrides.

The tube heat exchangers, 24 U-shaped copper circulation tubes, and 7 hydrogen tubes (mesh with 0.5-micron pores) were vertically penetrated the hydride bed. These tubes (0.5-inch O.D.) were arranged 30 mm apart in a triangular pattern. For efficient utilization of reaction heat, the heat loss to the vessel and circulation tubes must be minimized; however, every U-shaped tube was independently fixed to the top of the tank to withstand the mechanical tension caused by the expansion of metal hydrides and to determine the circulation flow path. The copper separators dissipated the temperature distribution of the metal hydride. In this study, every circulation tube was connected serially as shown in Fig. 2. Three thermocouples (T.C. A, T.C. B, and T.C. C) were inserted into the metal hydride bed (Fig. 2). Their readings roughly corresponded to the temperature of the metal hydride near the inlet or upstream of the flow path (A), the metal hydride around the midstream (B), and the metal hydride near the outlet or downstream of the flow path (C). The tank was covered with foam insulation to prevent heat loss.

The metal hydride used in this study was an Mm-based alloy (Japan Metals and Chemicals Co., Ltd.) whose composition was customized to achieve an equilibrium pressure within the desired range. 50 kg of the milled metal hydride was placed in the tank. The void fraction of the metal hydride to the tank volume was 0.52. The ratio of the total amount of absorbed hydrogen during operation (5400 NL) to the metal hydride was 0.97 wt% while the whole amount of absorbed hydrogen ($H/M = 1$) is about 1.39 wt%. The metal hydride was activated with cycles of absorption at 1.1 MPaG with cold circulation (7 °C) and desorption at vacuum with hot circulation (60 °C). The metal hydride was activated by vacuuming or

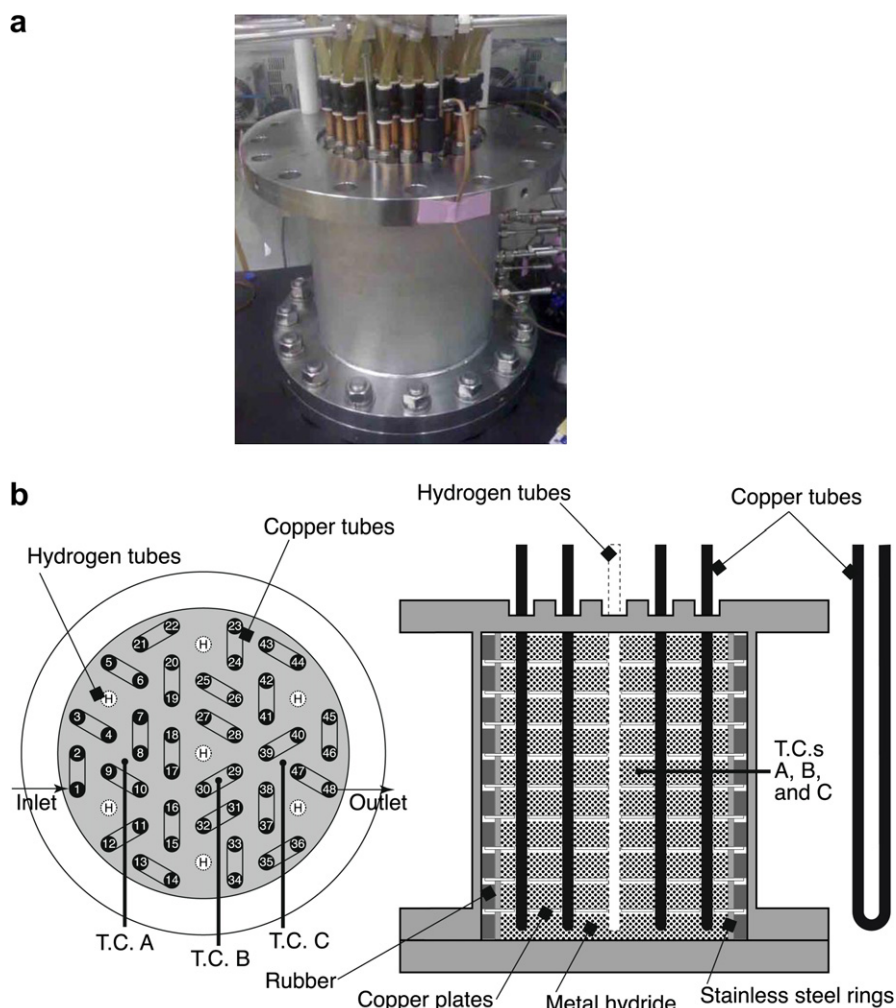


Fig. 2 – Schematic and photo of the metal hydride tank.

releasing hydrogen with hot water circulation and by pressurizing with cold water circulation. First, the metal hydride tank was vacuumed at the level of 20 Pa (abs.) using an oil rotary vacuum pump with 60 °C water circulation. Second the metal hydride tank was pressurized at the level of 1.1 MPa (abs.) with 7 °C water circulation. Then the similar vacuuming and pressurizing were repeated. After the absorption were identified by a mass flow meter, releasing hydrogen to outdoor without the vacuum pump and the similar pressurizing were conducted three times to complete the activation of metal hydride. During experiments in this study more than 10 cycles of absorption and desorption were conducted and the drift in hydrogen absorption ability was negligibly small.

2.2. Flow loops for hydrogen and circulation water

Fig. 3 schematically illustrates the flow loops for hydrogen and circulation water. A mass flow controller (KOFLOC MODEL 5100 (Full Scale 100 SLM) and KOFLOC CR-700) measured and controlled the hydrogen flow to an accuracy of 0.1 NL/min. The specially designed flow path enabled the use of the same mass flow controller for both the absorption and desorption processes to avoid introducing error due to variability in mass

flow controllers. The dead volume of the hydrogen flow loop was approximately less than 2 L and negligibly small compared to the entire absorbing hydrogen volume. Back-pressure valves at both ends of mass flow controller maintained the pressure drop through the mass flow controller. During the absorption process, hydrogen was introduced from hydrogen cylinders at 1.1 MPa (abs.) to emulate the operating pressure of a water electrolyzer. The purity of hydrogen used in this study was 99.99999%, and poisoning of the metal hydrides and resulting degradation were not observed.

The temperature of the circulation water was maintained by a thermostat controller (Tokyo Rikakikai Co., Ltd., NCC-2420) and the mass flow rate by a mass flow meter (Keyence, FD-SS2A) with a needle valve with an accuracy of 0.01 L/min. During either the hot or cold thermal energy recovery period, the thermostat controller unit with a pump controlled the temperature of the circulation water at the inlet and circulated the water. During either the self-heating or self-cooling period, another pump without heater or cooler circulated the water and the water temperature was changed only by the reaction heat of metal hydride. Room temperature was maintained at 22 °C to easily estimate the heat dissipation into the environment.

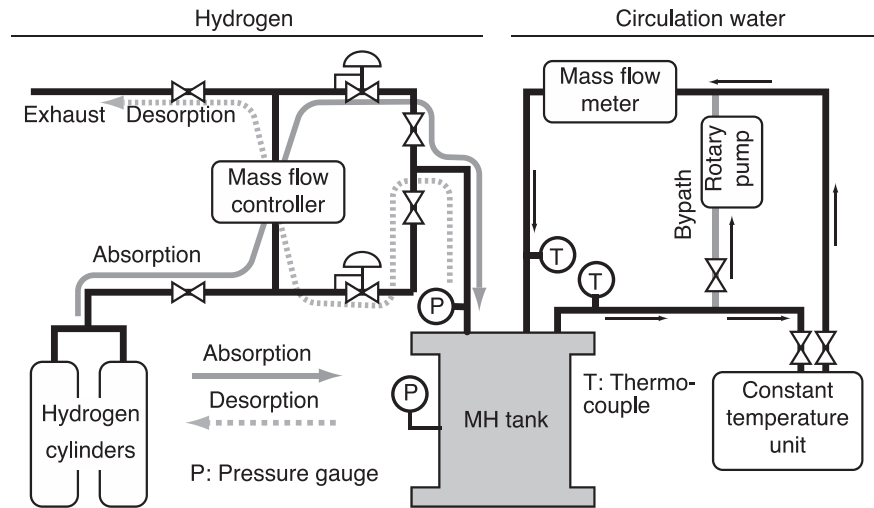


Fig. 3 – Schematic of the flow loops of hydrogen and circulation water.

3. Load leveling operation during a 24-h cycle

The experiments involved a 24-h cycle that emulated an ideal situation of load leveling to examine the storage ability of the metal hydride tank. In total, 4 cycles (4 days) were conducted to evaluate the effects of the initial state of the tank. Separate experiments involved (1) starting the absorption process from 32 °C uniform metal hydride temperature condition without a self-heating period and (2) starting the desorption process from 12 °C uniform metal hydride temperature condition without a self-heating period.

3.1. Operating procedure during the 24-h cycle experiments

Table 2 summarizes the operating conditions for the 24-h cycle. As specified in Section 2.1, the hydrogen absorption process was 9 h and desorption process was 13 h. The total storage volume of the metal hydride tank was 5400 NL, and thus the flow rate of hydrogen was 10 NL/min during the absorption process and 6.9 NL/min during desorption process. Each step of the operation during the 24-cycle is described below.

3.1.1. Initial state and absorption on day 1

Initial state of the metal hydride tank is specified with a composition of the metal hydride and temperature. The composition of metal hydride, F [H/M], is defined as the ratio

of absorbed hydrogen molecules to metal hydride molecules. As the direct estimation of F is difficult, we calculate F from total amount of hydrogen absorbed from the reference of composition, F_{ref} . The state of $F = 0$ can be achieved by having the metal hydride tank at vacuum. Pumping this amount of hydrogen, however, would damage a vacuum pump. We therefore defined the equilibrium state at 32 °C and atmospheric pressure, $F = 0.025$, as the reference of the composition, F_{ref} . This state for the metal hydride in the tank was easily obtained by circulating coolant at 32 °C and releasing hydrogen from the tank to outdoor without back pressure gauge. To make the initial state of the cycle experiment, $F = 0.18$, from the reference composition, F_{ref} , 1200 NL of hydrogen was supplied and the metal hydride bed reached a uniform temperature of 32 °C by coolant circulation. Pressure at the starting state was confirmed to be identical to the equilibrium pressure of the metal hydride as given by the manufacturer.

During absorption process, hydrogen was supplied at 10 NL/min to the tank. Reaction heat increased the temperature of both the metal hydride itself and the coolant. Because the temperature of the metal hydride and the circulation water was higher than 32 °C during absorption process on day 1, the temperature control unit supplied coolant at 32 °C at 1.0 NL/min into the inlet to remove the reaction heat during all period of absorption process. The flow rate of the circulation water, $G_c = 1.0$ L/min, was calculated from the required temperature gap and hydrogen flow rate using Eq. (1).

3.1.2. Interval after absorption

During the 1-h interval between the absorption process and subsequent desorption process, the inlet temperature of the coolant was set to 32 °C and the flow rate to 1.0 L/min to cool the metal hydride. Also during this interval, if a reversible fuel cell is used as a water electrolyzer and a fuel cell, the operating mode of the unitized fuel cell can be switched.

3.1.3. Desorption

Just after the 1-h interval, the metal hydride tanks released hydrogen at a constant flow rate of 6.9 NL/min via the mass

Table 2 – Operating procedure for load leveling.

Time[h]	0 (9 h)	9 (1 h)	10 (13 h)	23 (1 h)	24
Phase	Absorption	Int.	Desorption	Int.	
G_{H_2} [NL/min]	10	0	–6.9	0	
$T_{out}[C]$	$< = 32 > 32$		$> = 12 < 12$		
G_{water} [L/min]	1.0	1.0	1.0	0.41	1.0
$T_{in}[C]$	T_{out}	32	32	T_{out}	12

flow controller. During the desorption process, the endothermic reaction heat process chilled the metal hydride. The temperature of the metal hydride was initially higher than 12 °C, however, and the coolant temperature was not adequate to supply cold heat to the refrigeration system. Until the output temperature of the coolant fell below 12 °C, outflow of the coolant was not fed through the temperature control unit, but was directly fed back to the inlet via a bypass by using another pump to cool the metal hydride by the reaction heat itself. This self-cooling continued until the outlet temperature was lower than 12 °C (by monitoring the outlet temperature). The flow rate during this self-cooling was arbitrarily set at 1.0 L/min.

When the outlet temperature decreased to less than 12 °C, the reaction cold heat was utilized and recovered to support the refrigeration system. The circulation water was again fed into the temperature control unit, and the inlet temperature was maintained at 12 °C. The flow rate of the circulation water was reduced to 0.41 L/min to increase the temperature gap between the inlet and outlet.

3.1.4. Interval after desorption

During the 1-h interval between the desorption process and subsequent absorption process, the metal hydride was still below 12 °C and cold heat was recovered with the circulation water at a flow rate of 1.0 L/min.

3.1.5. Absorption after desorption and subsequent interval

The absorption process after the preceding desorption and interval required self-heating of the cold metal hydride, in contrast to the first absorption where temperature needed to be controlled from the onset. Using the bypass and the pump as in the self-cooling, the exothermic reaction heat was used to heat the metal hydride itself. After the outlet temperature exceeded 32 °C, the coolant was fed through the temperature control unit. The inlet temperature was maintained at 32 °C independently of the outlet temperature. The flow rate of hydrogen and the circulation water was 10 NL/min and 1.0 NL/min respectively.

3.2. Pressure and temperature profiles in the metal hydride tank

Fig. 4 shows representative results of the hydrogen flow rate and pressure during the 24-h cycle operation (day 3 is shown). A positive flow rate denotes absorption of hydrogen whereas a negative flow rate denotes desorption. In both processes, the hydrogen flow satisfied the regulated constant flow rate. Pressure of the tank also was within the pressure regulation range described in Section 1.2. The spikes at both intervals of the flow rate (circled in the figure) were due to the sudden pressure change in the mass flow controller when the flow path was changed during the 1-h interval. The accumulated flow rate was not affected by these spikes.

Fig. 5 shows the pressure profile during continuous operation on a composition–pressure diagram. Similar charts, known as PCT curves, are used here as pressure composition isotherms of the metal hydrides. Equilibrium pressure of metal hydrides depends on its composition and temperature and has hysteresis between the absorption and desorption

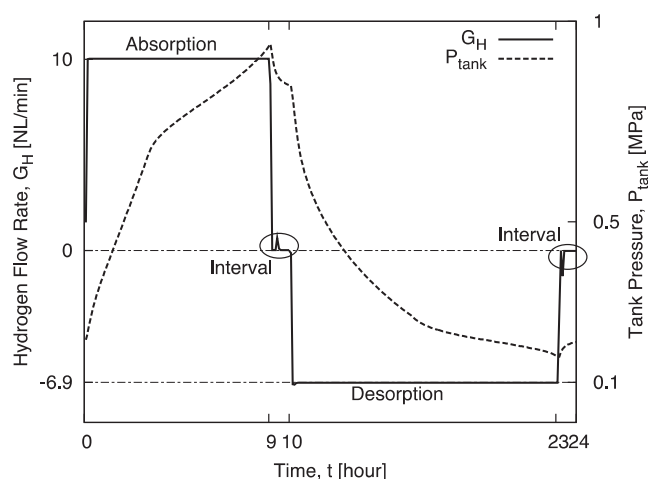


Fig. 4 – Hydrogen flow and tank pressure in 24-h experiment.

processes. The equilibrium pressure of the metal hydride at 32 °C and 12 °C stated by the alloy manufacturer is also plotted as thin dotted lines and small circles. Pressure profiles during separate experiments started at these points on the equilibrium pressure curves, indicating that the metal hydride temperature at the beginning of the process was uniformly maintained at a desired temperature, 32 °C or 12 °C. After the absorption/desorption process started, the pressure profiles departed from the equilibrium pressure curves; the pressure in absorption process became higher than the equilibrium pressure at 32 °C, and the pressure in desorption process became lower than the equilibrium pressure at 12 °C. These

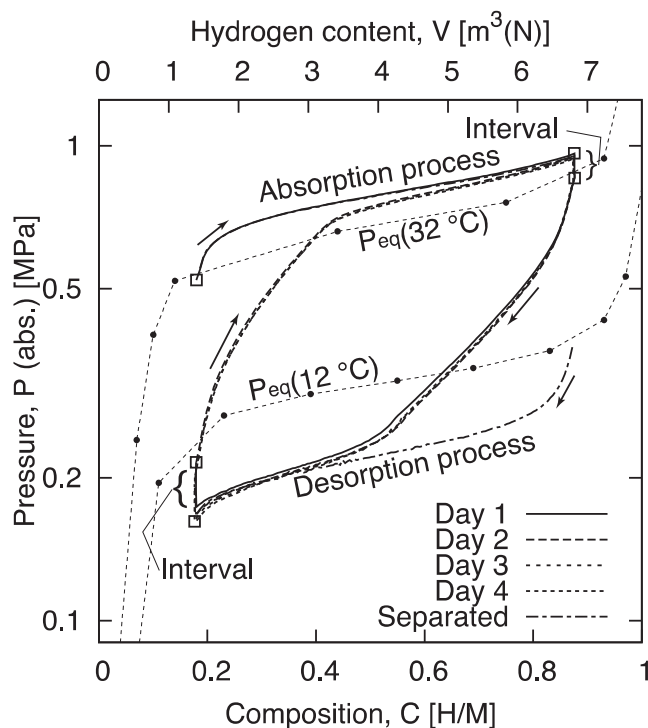


Fig. 5 – Tank pressure and composition of metal hydride during 24-h experiment and separated experiment.

pressure differences resulted from the temperature change of the metal hydride due to its reaction heat. The pressure difference from the equilibrium, however, did not grow steadily because the coolant removed the reaction heat and prevented the temperature change of the metal hydride. The average temperature of the metal hydride approached to the temperature at which the amount of the exchanged heat balanced the amount of the reaction heat.

The PCT curve for the 24-h cycle experiment shown in Fig. 5 (From Day 1 to Day 4) differed from those for separate experiments. The absorption process on day 1 started from a uniform metal hydride temperature and showed a curve identical to the curve for separated absorption experiments. After the absorption process on day 1 the pressure profile showed a closed loop; the end point of absorption and the 1-h interval is the starting point of desorption, and vice versa. The desorption process in the 24-h cycle experiment begins from a higher temperature and the absorption process in the 24-h cycle experiment begins from a lower temperature compared with the separated operation experiments. After the absorption process of day 2, the pressure profiles almost overlapped each other and the effects of the initial state disappear. The maximum and minimum pressures during the 24-h cycle experiment fell inside the range as described in Section 1.3. The initial pressures of the separated processes were equal to the equilibrium pressure because the temperature of the metal hydride and the tank were relatively uniform. The absorption/desorption process heats/cools the metal hydride and tank, and thus the pressure departs from the equilibrium pressure.

Fig. 6 shows the measured temperature profiles as described in Section 2.1. Just after the onset of absorption or desorption, the metal hydride temperature departs from the circulation temperature due to the reaction heat of the metal hydride. The temperature difference from the inlet circulation temperature increases from the upstream side (T.C. A) to the downstream side (T.C. C). The upstream side temperature increases and then decreases, indicating that the respective absorption/desorption process was almost completed at the upstream region because the heat exchange ability was higher near the inlet. The main part of the reaction might have moved to the downstream side. During the self-heating and

the self-cooling processes, the coolant exchanged heat between the upstream and downstream, thus decreasing the temperature difference between the upstream side (T.C. A) and the downstream side (T.C. C).

3.3. Recovering the heat source

Fig. 7 shows temperature profiles of the circulation water outlet, T_{out} . The 24-h cycle experiments required a period of self-heating and self-cooling before recovery of the thermal energy in both the absorption and desorption processes. Recovered thermal energies are defined as

$$Q_{heat} = (G_c p c)_c \int_{(T_{out} > 32^\circ \text{C})} (T_{out} - 32^\circ \text{C}) dt \quad (2)$$

$$Q_{cool} = G_c (p c)_c \int_{(T_{out} < 12^\circ \text{C})} (12^\circ \text{C} - T_{out}) dt \quad (3)$$

where Q_{heat} and Q_{cool} are the thermal energies recovered during the absorption process and the desorption process, respectively.

Table 3 shows the recovered thermal energy in both absorption and desorption processes. η means the ratio of recovered thermal energy to total reaction heat of absorption and desorption processes. Total reaction heat was 6.75 MJ (=5,400 NL $\times$ 1,250 J/NL, where 1250 J/NL is the reaction heat of the metal hydride per 1 NL as provided by the alloy manufacturer). The values of η for the separated experiments and that for absorption on day 1 were higher than the value of others because the separated experiments did not require significant heating or cooling of the metal hydride or tank. Recovered thermal energy on days 2, 3, and 4 showed similar values for η , indicating that 24 h is sufficient to eliminate the effects of the initial condition.

Table 4 shows the thermal capacity for each of the tank components. Components for the metal hydride tank had larger thermal capacities than the metal hydride itself, and thus heating and cooling the tank cannot be ignored. Although the reaction heat was exchanged via circulation water, sensible heat of circulation was dissipated to the tank through

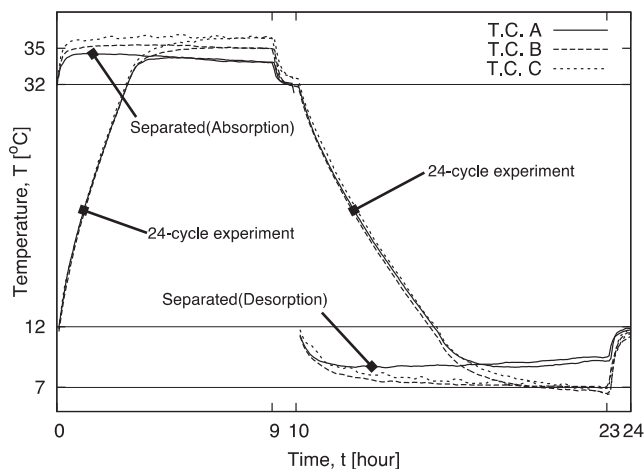


Fig. 6 – Temperature profiles in the MH tank.

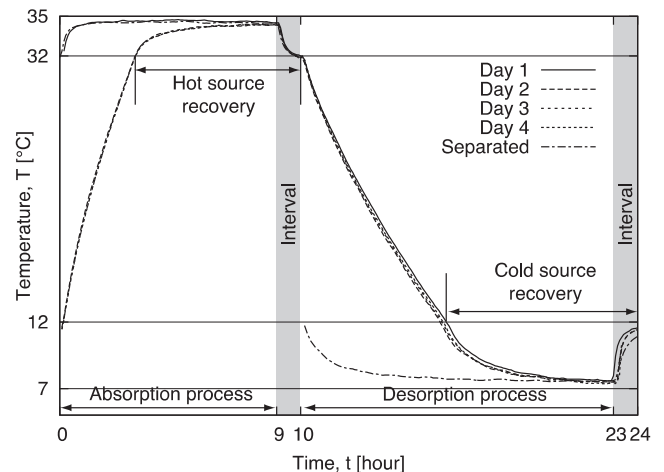


Fig. 7 – Temperature profiles at the circulation water outlet.

Table 3 – Recovered thermal energy.

	Absorption	Desorption
	Q[MJ] (η [-])	Q[MJ] (η [-])
Separated	5.72 (84.7%)	5.56 (82.3%) ^a
Day 1	5.94 (88.0%)	2.77 (41.1%)
Day 2	3.09 (45.8%)	2.98 (44.2%)
Day 3	3.14 (46.5%)	2.99 (44.3%)
Day 4	3.09 (45.9%)	2.77 (41.0%) ^b
Average	46.1% (Days 2, 3, and 4)	43.2% (Days 1, 2, and 3)

a Gc after desorption is 0.41 L/min.

b Cold source recovery process after desorption was missed.

the fittings of the circulation tubes. This dissipated thermal energy can be roughly estimated as the heat for warming or cooling of the tank. Heat balance was adequate, and thus the heat loss to the environment was not significant.

3.4. High hydrogen flow rate operation

Constant flow rates for both absorption and desorption processes were satisfied as shown in Section 3.3, indicating that the tank has the potential to absorb and desorb hydrogen at a higher flow rate. Experiments were then conducted using a 200% higher hydrogen flow rate (20 NL/min in absorption; 13.8 NL/min in desorption) and 400% higher (40 NL/min in absorption; 27.6 NL/min in desorption) to evaluate the limit of the heat transfer characteristics of the present design of the tank. The total amount of hydrogen storage was kept at 5400 NL. To maintain a thermal energy balance with these higher hydrogen flow rates, the coolant flow rate of each process was doubled or quadrupled. As well as 100% flow rate condition (10 NL/min in absorption; 6.9 NL/min in desorption), composition of the metal hydride at the onset of absorption was 0.18 and the temperature was 32 °C, whereas at the onset of desorption they were 0.88 and 12 °C, respectively.

Fig. 8 shows the hydrogen flow rate profile for each hydrogen flow rate. Although the 100% flow rate (10 NL/min in absorption; 6.9 NL/min in desorption) and 200% flow rate satisfied the constant flow rate requirements, the 400% flow rate did not. Under constant flow conditions, only 95% of the total hydrogen volume was absorbed and 65% desorbed.

A decrease in flow rate is caused by a pressure difference between the tank and mass flow controller. Fig. 9 shows the pressure increase during the absorption process and the pressure decrease during the desorption process for the 400% flow rate experiment. The pressure in the tank reached the limit even though the thermal balance of reaction heat and

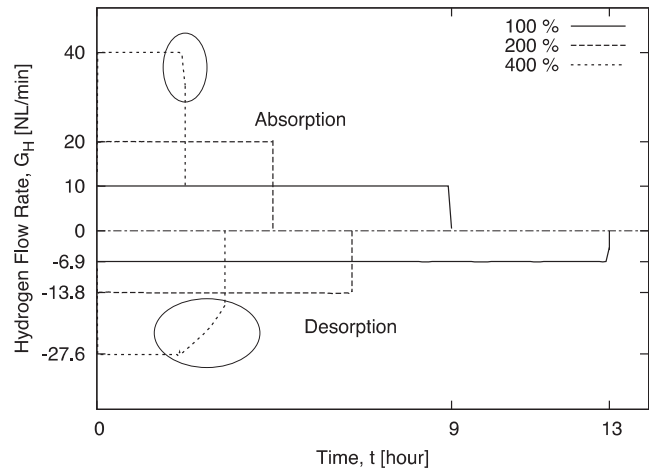


Fig. 8 – Flow rate profiles in the MH tank at different flow rates.

coolant was same regardless of flow rate. The reaction heat could not be removed under the higher flow rate conditions, because the heat exchange using the circulation tubes was insufficient to maintain the metal hydride temperature. Fig. 10 shows the outlet temperature for each flow rate condition. The temperature profile during the desorption process under the 200% flow rate condition might not visible in this figure due to overlapping with the profile for the 100% flow rate condition. Every profile approaches the outlet temperature calculated from thermal balance. However, under the 400% flow rate condition, the reaction heat could not be removed

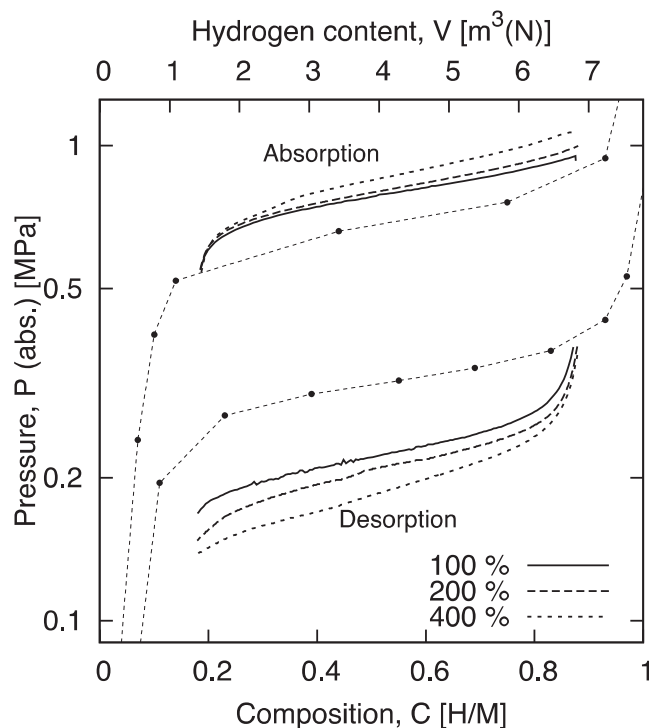


Fig. 9 – Tank pressure and composition of metal hydride during for different flow rates.

Table 4 – Thermal capacity of tank components.

Component	Material	Mass (kg)	Thermal capacity (kJ/K)
Metal hydride	AB ₅	50.0	20.5
Vessel	SUS304	148.0	87.3
Separators	Copper	19.3	7.3
Inner rings	SUS304	12.0	7.1
Rubbers	Silicon rubber	3.8	5.8
Total		233.1	128.0

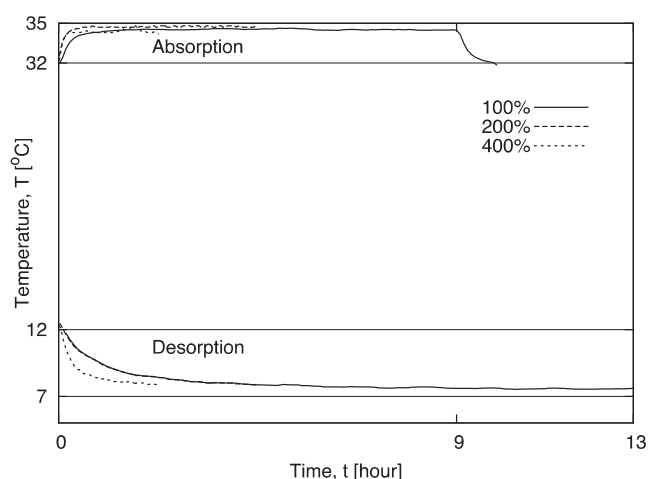


Fig. 10 – Outlet temperature during high hydrogen flow rate operation.

sufficiently due to the limit of the ability of the heat exchanger of the metal hydride tank.

4. Conclusion

An on-site energy storage system called a Totalized Hydrogen Energy Utilization System (THEUS) that uses metal hydride has been proposed to balance the energy demand gap between daytime and nighttime that affects electric power suppliers. A special operating procedure for this system has been designed to maximize the utilization of both hydrogen and reaction heat. A feasibility study demonstrated that the system satisfies flow rate, pressure, and temperature requirements for load leveling operation. The approximate amount of recovered cold source calculated from the thermal capacity of the tank system was about 43.2% of the total reaction heat. Experimental results in which the flow rate of hydrogen was doubled indicate that the tank system can be operated at higher flow rate.

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